

Typed topology request

fedctl submit run
--supernodes rpi4=10
--supernodes rpi5=10

deploy.supernodes

rpi4: 10

rpi5: 10

command override or config default

Typed client set

$C = C4 \text{ union } C5$

logical clients before physical placement



C4: rpi4



C5: rpi5

Placement map

$p : C \rightarrow H \times D$

- host may be concrete
- device class is preserved

Pinned host
or typed pool

node.unique.id

meta.device_type

Paired runtime role

one logical client becomes two matched roles

SuperNode

client-app

same index

DEVICE_TYPE

partition-id

worker pools

rpi4 host pool

rpi5 host pool

Device-labelled diagnostics

train time

update share

aggregate weight

staleness